

MATHEMATICS-II

Unit-III: Groups

Comprehensive Study Notes

Topic 1: Definition of Group

1.1 What is a Group?

A **group** is a set G together with a binary operation $*$ that satisfies four specific conditions, called the **group axioms**.

1.2 Group Axioms

A set G with a binary operation $*$ forms a group if it satisfies:

Axiom	Description	Mathematical Form
1. Closure	The operation on any two elements produces another element of the set	For all $a, b \in G$, $a * b \in G$
2. Associativity	The operation is associative	For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$
3. Identity	There exists an identity element	There exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$
4. Inverses	Every element has an inverse	For each $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

Group Axioms: Closure + Associativity + Identity + Inverses

1.3 Notation

- Operation can be denoted as $*$, $\cdot$, or just juxtaposition ab
- **Identity element** is usually denoted by e or 0 (for additive groups)
- **Inverse** of a is denoted by a^{-1} (multiplicative) or $-a$ (additive)

1.4 Abelian Group

If the group also satisfies the **commutative law**, it is called a **commutative** or **abelian group**.

$$a * b = b * a \text{ for all } a, b \in G$$

Topic 2: Groups of Numbers

2.1 Integers under Addition

Set: $\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$

Operation: Addition (+)

Axiom	Verification
Closure	Sum of two integers is an integer ✓
Associativity	$(a+b)+c = a+(b+c)$ ✓
Identity	0 is the identity: $a+0 = 0+a = a$ ✓
Inverses	Inverse of a is $-a$: $a + (-a) = 0$ ✓

Conclusion: $(\mathbb{Z}, +)$ is an **abelian group**.

2.2 Non-Zero Reals under Multiplication

Set: $\mathbb{R}^* = \mathbb{R} - \{0\}$

Operation: Multiplication ($\times$)

Axiom	Verification
Closure	Product of non-zero reals is non-zero real ✓
Associativity	$(a \times b) \times c = a \times (b \times c)$ ✓
Identity	1 is the identity: $a \times 1 = 1 \times a = a$ ✓
Inverses	Inverse of a is $1/a$: $a \times (1/a) = 1$ ✓

Conclusion: $(\mathbb{R}^*, \times)$ is an **abelian group**.

2.3 Integers under Multiplication

Axiom	Verification
Closure	Product of integers is integer ✓
Associativity	Holds ✓
Identity	1 is the identity ✓
Inverses	Only 1 and -1 have inverses in $\mathbb{Z}$

Conclusion: $(\mathbb{Z}, \times)$ is **NOT** a group.

2.4 Rational Numbers under Addition

Set: $\mathbb{Q}$

Operation: Addition (+)

Conclusion: $(\mathbb{Q}, +)$ is an **abelian group** (similar to integers).

2.5 Non-Zero Rationals under Multiplication

Set: $\mathbb{Q}^* = \mathbb{Q} - \{0\}$

Operation: Multiplication ($\times$)

Conclusion: $(\mathbb{Q}^*, \times)$ is an **abelian group**.

2.6 Summary: Groups of Numbers

Set	Operation	Group?	Type
$\mathbb{Z}$	Addition	Yes	Abelian
$\mathbb{Z}$	Multiplication	No	—
$\mathbb{Z} - \{0\}$	Multiplication	No	—

$\mathbb{Q}$	Addition	Yes	Abelian
$\mathbb{Q}^*$	Multiplication	Yes	Abelian
$\mathbb{R}$	Addition	Yes	Abelian
$\mathbb{R}^*$	Multiplication	Yes	Abelian

Topic 3: Groups of Residues

3.1 Residue Classes Modulo n

For a positive integer n , the set of residue classes modulo n is:

$$\mathbb{Z}_n = \{ \bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1} \}$$

Where $\bar{a}$ represents all integers congruent to a modulo n .

3.2 $(\mathbb{Z}_n, +)$ under Addition Modulo n

Set: $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$

Operation: Addition modulo n : $a \oplus b = (a+b) \bmod n$

Axiom	Verification
Closure	Sum modulo n is in $\mathbb{Z}_n$ ✓
Associativity	Follows from integer addition ✓
Identity	0 is the identity ✓
Inverses	Inverse of a is $n-a$ (for $a \neq 0$) ✓

Conclusion: $(\mathbb{Z}_n, +)$ is an **abelian group**.

Example: $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

3.3 $(\mathbb{Z}_n^*, \times)$ under Multiplication Modulo n

Set: $\mathbb{Z}_n^* = \{ a \in \mathbb{Z}_n \mid \gcd(a, n) = 1 \}$ (units modulo n)

Operation: Multiplication modulo n

Axiom**Verification**

Closure	Product of units is a unit ✓
Associativity	Follows from integer multiplication ✓
Identity	1 is the identity ✓
Inverses	Every unit has a multiplicative inverse modulo n ✓

Conclusion: $(\mathbb{Z}_n^*, \times)$ is an **abelian group**.

Example: $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$

Order of $\mathbb{Z}_n^*$: $|\mathbb{Z}_n^*| = \phi(n)$ (Euler's phi function)

3.4 Solved Examples

Example 1: Check if $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ is a group under addition modulo 4.

Step 1: Closure – $2+3 = 5 \equiv 1 \pmod{4} \in \mathbb{Z}_4$ ✓ All sums are in $\mathbb{Z}_4$.

Step 2: Associativity – Holds ✓

Step 3: Identity – 0 is the identity ✓

Step 4: Inverses

$$0^{-1} = 0, 1^{-1} = 3, 2^{-1} = 2, 3^{-1} = 1$$

Conclusion: $(\mathbb{Z}_4, +)$ is an abelian group ✓

Example 2: Check if $\mathbb{Z}_5^* = \{1, 2, 3, 4\}$ is a group under multiplication modulo 5.

Step 1: Closure – $2 \times 3 = 6 \equiv 1 \pmod{5}$ ✓; $3 \times 4 = 12 \equiv 2 \pmod{5}$ ✓

Step 2: Associativity – Holds ✓

Step 3: Identity – 1 is the identity ✓

Step 4: Inverses

$$1^{-1} = 1, 2^{-1} = 3, 3^{-1} = 2, 4^{-1} = 4$$

Conclusion: $(\mathbb{Z}_5^*, \times)$ is an abelian group ✓

Example 3: Check if $\mathbb{Z}_6^* = \{1, 5\}$ is a group under multiplication modulo 6.

Step 1: Closure – $1 \times 5 = 5 \in \mathbb{Z}_6^*$ ✓; $5 \times 5 = 25 \equiv 1 \pmod{6}$ ✓

Step 2: Associativity – Holds ✓

Step 3: Identity – 1 is the identity ✓

Step 4: Inverses

$$1^{-1} = 1, 5^{-1} = 5$$

Conclusion: $(\mathbb{Z}_6^*, \times)$ is an abelian group ✓

Topic 4: Groups of Matrices

4.1 Matrix Groups

Many groups have matrices as their elements. The operation is usually either **matrix addition** or **matrix multiplication**.

4.2 $M_{m \times n}(\mathbb{R})$ under Matrix Addition

Set: All $m \times n$ matrices with real entries

Operation: Matrix addition

Axiom	Verification
Closure	Sum of two $m \times n$ matrices is an $m \times n$ matrix ✓
Associativity	Matrix addition is associative ✓
Identity	Zero matrix (all entries 0) is the identity ✓
Inverses	Inverse of matrix A is $-A$ (negate all entries) ✓

Conclusion: $(M_{m \times n}(\mathbb{R}), +)$ is an **abelian group**.

4.3 $GL_n(\mathbb{R})$ – General Linear Group

Set: All invertible $n \times n$ matrices with real entries

$$GL_n(\mathbb{R}) = \{ A \in M_{n \times n}(\mathbb{R}) \mid \det(A) \neq 0 \}$$

Operation: Matrix multiplication

Axiom	Verification
-------	--------------

Closure	Product of invertible matrices is invertible ✓
Associativity	Matrix multiplication is associative ✓
Identity	Identity matrix I_n is the identity ✓
Inverses	Every invertible matrix has an inverse ✓

Conclusion: $(GL_n(\mathbb{R}), \times)$ is a **group** (not abelian for $n \geq 2$).

4.4 $SL_n(\mathbb{R})$ – Special Linear Group

Set: All $n \times n$ matrices with determinant 1

$$SL_n(\mathbb{R}) = \{ A \in GL_n(\mathbb{R}) \mid \det(A) = 1 \}$$

Operation: Matrix multiplication

Conclusion: $SL_n(\mathbb{R})$ is a **subgroup** of $GL_n(\mathbb{R})$.

4.5 Solved Examples

Example 1: Show that 2×2 matrices with real entries form a group under matrix addition.

Solution:

Step 1: Closure

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

This is also a 2×2 matrix with real entries ✓

Step 2: Associativity – Matrix addition is associative ✓

Step 3: Identity – Zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is the identity ✓

Step 4: Inverses – Inverse is $-\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$ ✓

Conclusion: The set forms an abelian group under matrix addition ✓

Example 2: Check if 2×2 diagonal matrices form a group under matrix addition.

Set: $G = \{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{R} \}$

Solution:

Closure: Sum of diagonal matrices is diagonal ✓

Associativity: Holds ✓

Identity: Zero matrix is diagonal ✓

Inverses: Negative of diagonal matrix is diagonal ✓

Conclusion: Yes, it is an abelian group under addition.

Topic 5: Groups of Functions

5.1 Functions under Addition

Set: All real-valued functions with domain $\mathbb{R}$

Operation: Pointwise addition: $(f+g)(x) = f(x) + g(x)$

Axiom	Verification
Closure	Sum of functions is a function ✓
Associativity	Function addition is associative ✓
Identity	Zero function $f(x) = 0$ for all x is the identity ✓
Inverses	Inverse of f is $-f$ defined by $(-f)(x) = -f(x)$ ✓

Conclusion: This is an **abelian group**.

5.2 Non-Zero Functions under Multiplication

Condition: Functions that are **nowhere zero** (i.e., $f(x) \neq 0$ for all x)

Operation: Pointwise multiplication: $(f \times g)(x) = f(x) \times g(x)$

Axiom	Verification
Closure	Product of nowhere-zero functions is nowhere-zero ✓
Associativity	Holds ✓
Identity	Constant function $f(x) = 1$ is the identity ✓
Inverses	Inverse of f is $1/f$, defined by $(1/f)(x) = 1/f(x)$ ✓

Conclusion: This is an **abelian group**.

Important: If "non-zero" means "not the zero function," then it is **not** a group because a function like $f(x) = x$ does not have an everywhere-defined inverse.

5.3 Solved Examples

Example 1: Show that real-valued functions form an abelian group under addition.

Solution:

Step 1: Closure – If f and g are functions, $f+g$ is a function ✓

Step 2: Associativity – $(f+g)+h = f+(g+h)$ ✓

Step 3: Identity – Zero function $z(x) = 0$ is the identity ✓

Step 4: Inverses – For each f , $-f$ defined by $(-f)(x) = -f(x)$ is the inverse ✓

Step 5: Commutativity – $f+g = g+f$ ✓

Conclusion: The set is an abelian group under addition ✓

Example 2: Why do non-zero functions NOT form a group under multiplication?

Explanation: If "non-zero" means "not the zero function", then the function $f(x) = x$ has no multiplicative inverse.

The inverse would need to be $1/x$, which is not defined at $x = 0$.

Therefore, the set does not have inverses for all elements.

Conclusion: Not a group.

But: If "non-zero" means "nowhere zero" ($f(x) \neq 0$ for all x), then it **is** a group ✓

Topic 6: Groups of Subsets of a Set

6.1 Power Set and Symmetric Difference

Set: $P(S)$ = all subsets of a set S

Operation: Symmetric difference defined as:

$$A * B = (A \cup B) - (A \cap B)$$

(The set of elements in exactly one of A and B)

6.2 The Group $(P(S), *)$

Claim: $P(S)$ with symmetric difference forms an abelian group.

Axiom	Verification
Closure	$A * B$ is a subset of S ✓
Associativity	Symmetric difference is associative ✓
Identity	Empty set $\emptyset$ is the identity: $A * \emptyset = A$ ✓
Inverses	Each element is its own inverse: $A * A = \emptyset$ ✓
Commutativity	$A * B = B * A$ ✓

Conclusion: $(P(S), *)$ is an **abelian group**.

6.3 Solved Examples

Example 1: Show that $P(S)$ with symmetric difference is a group.

Solution:

Step 1: Closure – For any subsets $A, B \subseteq S$, $A * B = (A \cup B) - (A \cap B)$ is also a subset of S ✓

Step 2: Associativity – Given ✓

Step 3: Identity – $\emptyset$ is the identity because $A * \emptyset = (A \cup \emptyset) - (A \cap \emptyset) = A - \emptyset = A$ ✓

Step 4: Inverses – Each element is self-inverse because $A * A = (A \cup A) - (A \cap A) = A - A = \emptyset$ ✓

Conclusion: $(P(S), *)$ is an abelian group ✓

Example 2: For $S = \{1,2,3\}$, solve $\{1,2,4\} * X = \{3,4\}$.

Solution:

Since each element is self-inverse: $X = \{1,2,4\} * \{3,4\}$

$$X = \{1,2,4\} \cup \{3,4\} - \{1,2,4\} \cap \{3,4\}$$

$$X = \{1,2,3,4\} - \{4\} = \{1,2,3\}$$

Check: $\{1,2,4\} * \{1,2,3\} = \{3,4\}$ ✓

Topic 7: Properties of Groups

7.1 Basic Properties

Property	Description	Mathematical Form
Uniqueness of Identity	Identity element is unique	If e_1 and e_2 are identities, $e_1 = e_2$
Uniqueness of Inverses	Each element has a unique inverse	If $a*b = e$ and $a*c = e$, then $b = c$
Cancellation Law	Left and right cancellation	$ab = ac \Rightarrow b = c$
Inverse of Inverse	Inverse of inverse is the element	$(a^{-1})^{-1} = a$
Inverse of Product	Inverse of product	$(ab)^{-1} = b^{-1}a^{-1}$

7.2 Order of an Element

Definition: The order of an element a in a group G is the smallest positive integer n such that:

$$a^n = e$$

If no such n exists, a has infinite order. **Notation:** $o(a)$ or $|a|$.

7.3 Order of a Group

Definition: The order of a group G is the number of elements in G . **Notation:** $|G|$

7.4 Subgroups

Definition: A subset H of a group G is a **subgroup** if H is itself a group under the same operation.

Conditions for Subgroup:

1. $ab \in H$ for all $a, b \in H$
2. $e \in H$
3. $a^{-1} \in H$ for all $a \in H$

7.5 Lagrange's Theorem

Theorem: If H is a subgroup of a finite group G , then the order of H divides the order of G .

$$|H| \mid |G|$$

Corollaries:

1. For any $a \in G$, $o(a)$ divides $|G|$
2. For any $a \in G$, $a^{|G|} = e$
3. Any group of prime order is cyclic

7.6 Solved Examples

Example 1: Prove that identity element is unique.

Solution: Suppose e_1 and e_2 are both identity elements in group G .

$$e_1 = e_1 * e_2 \text{ (using } e_2 \text{ as identity)}$$

$$e_1 * e_2 = e_2 \text{ (using } e_1 \text{ as identity)}$$

$$\text{Therefore, } e_1 = e_2 \checkmark$$

Conclusion: Identity is unique.

Example 2: Prove that inverses are unique.

Solution: Suppose a has two inverses b and c .

$$a * b = e \text{ and } a * c = e$$

$$b = b * e = b * (a * c) = (b * a) * c = e * c = c$$

Therefore, $b = c$ ✓

Conclusion: Inverses are unique.

Example 3: Find the order of each element in $\mathbb{Z}_4$ under addition.

Solution:

0 has order 1 (since $0 = 0$)

1 has order 4 (since $1+1+1+1 = 4 \equiv 0$)

2 has order 2 (since $2+2 = 4 \equiv 0$)

3 has order 4 (since $3+3+3+3 = 12 \equiv 0$)

Example 4: Prove that $(ab)^{-1} = b^{-1}a^{-1}$.

Solution:

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = a(e)a^{-1} = aa^{-1} = e$$

$$\text{Similarly, } (b^{-1}a^{-1})(ab) = e$$

$$\text{Therefore, } (ab)^{-1} = b^{-1}a^{-1} \checkmark$$

Topic 8: Characterization of Groups

8.1 Alternative Characterization

A group can be characterized by:

1. **Closure** under the operation
2. **Associativity** of the operation
3. **Existence of identity**
4. **Existence of inverses**

8.2 Finite Groups vs Infinite Groups

Type	Description	Examples
Finite Group	Finite number of elements	$\mathbb{Z}_n$, Symmetric groups
Infinite Group	Infinite number of elements	$\mathbb{Z}$, $\mathbb{R}$

8.3 Abelian vs Non-Abelian Groups

Type	Description	Examples
Abelian Group	Commutative operation	$\mathbb{Z}$, $\mathbb{Z}_n$, $\mathbb{R}$, $\mathbb{Q}$
Non-Abelian Group	Not commutative	$\text{GL}_n(\mathbb{R})$ for $n \geq 2$

8.4 Cayley's Theorem

Theorem: Every group is isomorphic to a **permutation group**.

Topic 9: Cyclic Groups

9.1 Definition

A group G is **cyclic** if every element of G can be written as a power of a single element a .

$$G = \langle a \rangle = \{ a^n \mid n \in \mathbb{Z} \}$$

The element a is called a **generator** of G .

9.2 Properties of Cyclic Groups

Property	Description
Abelian	Every cyclic group is abelian
Subgroups	Every subgroup of a cyclic group is cyclic
Isomorphism	Infinite cyclic groups are isomorphic to $\mathbb{Z}$
Finite Order	Finite cyclic groups of order n are isomorphic to $\mathbb{Z}_n$

9.3 Generators of Cyclic Groups

For $G = \langle a \rangle$ of order n :

Condition	Generator
a^k generates G	$\gcd(k, n) = 1$
Number of generators	$\varphi(n)$ (Euler's phi function)

9.4 Examples of Cyclic Groups

Group	Generator	Order
$\mathbb{Z}$ (under +)	1 or -1	Infinite
$\mathbb{Z}_n$ (under +)	Any k with $\gcd(k, n) = 1$	n
$\mathbb{Z}_5^*$ (under $\times$)	2, 3	4
$\mathbb{Z}_7^*$ (under $\times$)	3, 5	6

9.5 Solved Examples

Example 1: Show that $\mathbb{Z}$ is cyclic.

Solution:

$$\mathbb{Z} = \langle 1 \rangle = \{ n \cdot 1 \mid n \in \mathbb{Z} \}$$

Every integer can be written as a multiple of 1.

Therefore, $\mathbb{Z}$ is cyclic with generator 1.

Also: $\mathbb{Z} = \langle -1 \rangle$

Example 2: Show that $\mathbb{Z}_4$ is cyclic under addition.

Solution:

1 generates all elements: $1^1=1, 1^2=2, 1^3=3, 1^4=0$

Therefore, $\mathbb{Z}_4 = \langle 1 \rangle$ is cyclic.

Also: 3 is also a generator: $3^1=3, 3^2=2, 3^3=1, 3^4=0$

Example 3: Find all generators of $\mathbb{Z}_6$.

Solution:

$\mathbb{Z}_6$ has order 6.

Generators are k such that $\gcd(k, 6) = 1$.

$\gcd(1,6)=1 \checkmark, \gcd(2,6)=2, \gcd(3,6)=3, \gcd(4,6)=2, \gcd(5,6)=1 \checkmark$

Generators: 1 and 5

Example 4: Find all generators of $\mathbb{Z}_7$.

Solution:

$\mathbb{Z}_7$ has order 7 (prime).

Generators are k such that $\gcd(k, 7) = 1$.

All 1,2,3,4,5,6 have $\gcd = 1$.

Generators: 1, 2, 3, 4, 5, 6

Unit-III: Important Questions

Short Answer Questions (2–5 Marks)

1. Define a group and state its axioms.
2. Define an abelian group.
3. What is a cyclic group? Give an example.
4. Define the order of an element in a group.
5. Define the order of a group.
6. State Lagrange's theorem.
7. What is a subgroup? Give conditions.
8. Give an example of a group of residues.
9. Give an example of a group of matrices.
10. Show that $(\mathbb{Z}_4, +)$ is a group.
11. What is the difference between abelian and non-abelian groups?
12. Define symmetric difference of two sets.
13. Show that every cyclic group is abelian.

Long Answer Questions (10 Marks)

1. Define a group. Verify the group axioms for: (a) Integers under addition (b) Non-zero reals under multiplication (c) Residue classes modulo n under addition
2. Explain groups of matrices with examples.
3. Explain groups of functions with examples.
4. Show that the set of all subsets of a set S with symmetric difference forms a group.
5. State and prove properties of groups: (a) Uniqueness of identity (b) Uniqueness of inverses (c) Cancellation laws
6. Define cyclic groups. Prove that: (a) Every cyclic group is abelian (b) Every subgroup of a cyclic group is cyclic
7. State and prove Lagrange's theorem.
8. Find all generators of: (a) $\mathbb{Z}_8$ (b) $\mathbb{Z}_{10}$ (c) $\mathbb{Z}_7$

Objective Questions (1 Mark)

1. The identity element in $(\mathbb{Z}, +)$ is:

- (a) 0 (b) 1 (c) -1 (d) 2

Answer: (a)

2. The inverse of 5 in $(\mathbb{Z}_7, +)$ is:

- (a) 2 (b) 3 (c) 5 (d) 7

Answer: (a)

3. $(\mathbb{Z}_5^*, \times)$ has order:

- (a) 2 (b) 3 (c) 4 (d) 5

Answer: (c)

4. The identity element for symmetric difference is:

- (a) S (b) $\emptyset$ (c) {e} (d) None

Answer: (b)

5. Every cyclic group is:

- (a) Abelian (b) Non-abelian (c) Finite (d) Infinite

Answer: (a)

6. The order of $\mathbb{Z}_n$ under addition is:

- (a) n (b) $n-1$ (c) $\phi(n)$ (d) 1

Answer: (a)

7. $GL_2(\mathbb{R})$ is:

- (a) Abelian (b) Non-abelian (c) Finite (d) None

Answer: (b)

8. In $(\mathbb{Z}_4, +)$, the order of 2 is:

- (a) 1 (b) 2 (c) 3 (d) 4

Answer: (b)

Quick Revision

Group Axioms Summary

Condition	Property
Closure	$a, b \in G \Rightarrow a*b \in G$
Associativity	$(a*b)*c = a*(b*c)$
Identity	$\exists e \in G$ such that $e*a = a*e = a$

Inverses	$\forall a \in G, \exists a^{-1}$ such that $a * a^{-1} = a^{-1} * a = e$
----------	---

Examples of Groups

Set	Operation	Group Type
$\mathbb{Z}$	Addition	Abelian
$\mathbb{Z}_n$	Addition mod n	Abelian
$\mathbb{Z}_n^*$	Multiplication mod n	Abelian
$\mathbb{R}^*$	Multiplication	Abelian
$GL_n(\mathbb{R})$	Matrix Multiplication	Non-abelian
$P(S)$	Symmetric Difference	Abelian
Functions	Addition	Abelian

Key Formulas

Formula	Description
$e * a = a * e = a$	Identity property
$a * a^{-1} = a^{-1} * a = e$	Inverse property
$(ab)^{-1} = b^{-1}a^{-1}$	Inverse of product
$ H \mid G $	Lagrange's theorem
$G = \langle a \rangle$	Cyclic group

End of Unit-III Notes